

Phases 4 and 5

Racing together, and the record books

BeamMP Race Manager | Build 0.5.8

Phase 4 turns one driver at a time into a field. Phase 5 keeps what they did. This is what is built and what each part does.

1 Races you make and join

A race is a thing you make, and others join.

1. Pick a course, laps, driving mode and class. Make the race **public** or **invite only**, choosing people from the player list.
2. Everyone who joins is placed on the grid. Nothing is armed. The start line does nothing, however many times it is crossed.
3. The host presses **Start Race**. Only then does the line go live.
4. Each driver's clock starts when they cross it, so the timing stays exact for everyone even if they roll off a second apart.

Only the host can start it. A race that has started refuses late joiners. If the host leaves before starting, the race passes to whoever joined first rather than dying. One race per course. Different courses at the same time are fine, and driving alone works exactly as before.

2 One grid, one result

Everyone is placed in a row behind the line rather than all on the same spot.

The results come out as one table in finishing order, with the gap to the leader and the gap to the car in front. Every row carries its splits, its best lap and the penalties it picked up, with a reason against each one.

Somebody who leaves half way does not hold up the people still driving. They are filed as a did not finish and the race carries on. The results wait for the last car still on track and then go to everybody at once.

3 The pit

A checkpoint can be marked as a **pit** when you build the course. It is not part of the lap and it is drawn in its own colour. While you are inside it:

- Fuel fills to 100% instead of a quarter.
- Repair and Spare tire add nothing to your race time.
- Re-rack Spares is offered, which is a pit job only.
- The holds still run in full, and the clock is still going. A stop costs the time it takes and nothing on top.

4 Speed zones

This is the 37 mph idea, between any two checkpoints you pick. A zone is a start gate, an end gate and a limit, and a course can have as many as needed.

The server reads the speeds it already collects, so nothing is trusted to a driver's own game. Going over draws a warning first. The 30 second penalty lands only after three seconds over, so one bump over a crest is free, and it will not charge twice until you have dropped back under.

Zones are set from the console:

```
rm zone baja-1000 12 14 37
rm zones baja-1000
rm zoneclear baja-1000
```

5 Teams

Two drivers, one car model, one result.

Either driver can send an invite or ask to join. Both sides have to be in the same vehicle before a team can start, and the model is read off the car itself rather than trusted to anyone's own game. If the cars stop matching, the team breaks up on its own and both drivers are told why.

A team result is the two corrected times added together. Both drivers have to finish for the team to place, so a team is only as quick as the pair of them. The individual rows stay as they are, and the team standing sits beside them.

A team lives for the evening. It is not an account and nothing is written to disk, so nobody has to leave a team they made last week.

6 The record books

Every course keeps a book per mode. The Records button shows them.

- Each driver's best corrected run, fastest first, with when it was set.
- The fastest clean lap anyone has driven, held separately.
- Your own position when you are outside the top ten.
- Controller and wheel are separate books, as agreed in phase 2.

The moment a race finishes the run goes into the books before the results go out, so the results screen hangs the badge on the row it belongs to: **course record**, a **class record**, or **personal best**. The books live on disk and survive restarts.

Two things never make the books. A run the server marked as suspect, because its clock could not be vouched for. And any lap with a cut checkpoint in it, which was already refused a best lap by the results.

7 Records by class

Your seventeen classes are in, as you sent them, in your order, under the two divisions you grouped them under. They are on the race screen as a row of buttons under Limited and Unlimited.

A class is entered rather than worked out from the car. You allow every vehicle, and two drivers in the same truck can be running different classes, so the car cannot tell the server which book a run belongs in. The driver picks a class before setting off, or whoever makes the race picks one for the

whole grid, which is the way your bot announces them. Picking nothing is allowed: the run still counts, it just sits outside the class books.

Your best is kept once per class, not once per course. A slower run in a second class is that class's first entry rather than a run that failed to beat you.

The record screen has a filter across the top: every class, or one of them. It only offers the classes that have actually been raced on that course, so it stays short.

The line each finished race writes to disk carries the class and the car, so your Discord bot can read both without being told anything new.

8 Qualifying

A course marked as qualifying now sets the grid.

Whoever was quickest in the qualifying session starts at the front of the next race on that course. Anyone who was not in it lines up behind those who were, in the order they joined, so a late arrival starts at the back rather than being turned away.

The order lasts the session. A grid earned this evening should not still be deciding places next week.

9 Experience and levels

Finishing pays by where you came. Two hundred for a win, two less for each place after it, never below twenty. A level is a thousand points.

The level sits next to the name in the player list and follows the account, so it survives a restart for the same reason the name does. Nothing is paid to a guest, who has no record to put it in, and nothing is paid for a run the clock could not vouch for.

Levels start at nothing, because there are no old results to pay out for. They only move from here.

10 Kick, ban and granting experience

An admin can remove somebody now, or remove them and turn them away next time. A ban follows the account rather than a name, which can be changed, or a session, which is handed out fresh on every join. The list is on disk, so a restart does not quietly let everybody back in.

Somebody banned is turned away as they knock, before anything is written down for them. In game nobody can act on somebody who ranks as high as they do, or on themselves.

Experience can be handed out or taken back by name, for a race run outside the mod or a result that needs putting right.

```
rm kick Marx wrecking
rm ban Marx wrecking
rm unban Marx
rm bans
rm xp Marx 500
rm xp Marx -200
```

11 The console

Everything you can set from the panel console, in one place.

```
rm status          plugin health, one line
rm players         who is on, level, speed, ping, role
```

rm races	runs on track now
rm tracks	saved courses
rm track <id>	checkpoints for one course
rm records <id> [class]	the board for a course, whole or one class
rm classes	the race classes, by division
rm gatewidth <course> <metres>	set the gate width on a course
rm zone <id> <from> <to> <mph>	add a speed zone
rm zones <id>	the speed zones on a course
rm zoneclear <id>	remove every zone from a course
rm role <name> <role>	owner, admin, staff or player
rm kick <name> [why]	remove somebody now
rm ban <name> [why]	remove them and keep them out
rm unban <name>	lift a ban
rm bans	who is banned, and why
rm xp <name> <amount>	hand out experience
rm save	write every store now

12 The numbers

Every part above is covered by tests that run without the game.

Suite	Tests
Phase 1, identity	91
Phase 2, racing	141
Phase 3, the bottom bar and the rack	68
Phase 4, lobby, pit, zones	71
Phase 5, the books	31
Teams, experience, bans, classes, qualifying	84
Results and the race log	120
Gate shape and putting a car back	50
Course capture and shape	31
The clock and the lap bar	34
Total	721

On top of those, 196 checks read the interface and the code together and stop the build when the two stop agreeing, plus a reader over every client file and a check that every key binding lands on something real.

The sums that matter were also tested by breaking them on purpose: charging at the wrong end, letting marked runs into the books, ignoring the class a driver entered, keeping one best per course instead of one per class, starting a grid without the class its race was made in. Every one was caught.

13 What you can set

Nothing below needs a new build.

- The classes, their order and their divisions.
- What a finish pays, and how much a level costs.
- Every penalty, and how long each pit job holds the car.
- Speed zones, their limits and the grace before a penalty.
- Gate widths on a saved course.

- Who is owner, admin, staff or player.
-

The build is **0.5.7** and the console prints it on every restart, so you can always tell which one you are running.

The record books start empty, so the first finished race on each course sets its first record.